Reg. No. | | | | | | | | | | | [13CE54]

FIFTH SEMESTER B.Tech/B.E EXAMINATION, DECEMEBER-2015

DESIGN OF RCC STRICTIRES

Time: 3 Hours **Maximum Marks:70**

Instructions: **1.** Answer any **FIVE** full questions.
2. Missing data may be suitably assumed, if any
3. Draw neat diagrams wherever needed
4. Use of IS 456-2000 and SP: 16-1980 are permitted

1. a. Explain the following **06**
- **i.** Characteristic load
- **ii.** Characteristic strength
- **iii.** Partial safety factors for loads and materials

b. Explain the basic principles of limit state method of RCC design. **04**

c. Differentiate working stress and limit state methods. **04**

2. a. Explain the following terms **06**
- **i.** Balanced section
- **ii.** under reinforced section
- **iii.** over reinforced section

b. Obtain stress Block parameters for limit state of collapse. And derive moment of resistance of section. **08**

3. a. Explain long term and short term deflections. **04**

b. A rectangular simply supported beam of effective span 5m is of size 300mmx700mm. It is reinforced with 4 bars of 18mm with an effective cover of 50mm determine the short term deflection due to an Imposed working load of 20KN|m. If $f_{cks} = 25|mm^2$ and $f_y = 415N|mm^2$. **10**

4. A cantilever beam of 4m span carries a load of 40KN|m (Factored Load). The width of the Beam is 230mm. Design the Beam for flexure and shear. Sketch the reinforcement details. Take M_{20} concrete and Fe 415 steel. **14**

5. Design a R.C slab for a room measuring 4mx5m from inside. The slab carries a live load of 2KN|m^2 and floor load of 0.5 KN|m^2. The slab is simply supported at all the four edges with corners free to lift. Use M-20 grade concrete and Fe-415 grade steel. Sketch the reinforcement details. **14**

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6. Design a short RC column is subjected to 600KN ultimate load and 100Kn-m ultimate bending moment. The effective length of column is 4.5m use M20 grade concrete and fe415 steel. Sketch the details. **14**

7. A square footing has to transfer ultimate load 1200KN from a square column of 500mmx500mm. Design the footing taking SBC of soil as 200KN|m^2 use M20 grade of concrete and Fe-415 steel. Sketch the reinforcement details. **14**

8. Design a Doglegged stair for the stair case hall of 3.1mx4.75m (clear dimensions) the floor to floor height is 3.0m .Take live load as 5KN|m^2. Use M20 grade concrete and Fe-415 steel. Sketch the reinforcement details. Given tread and rise of steps are 250mm and 150mm respectively. **14**